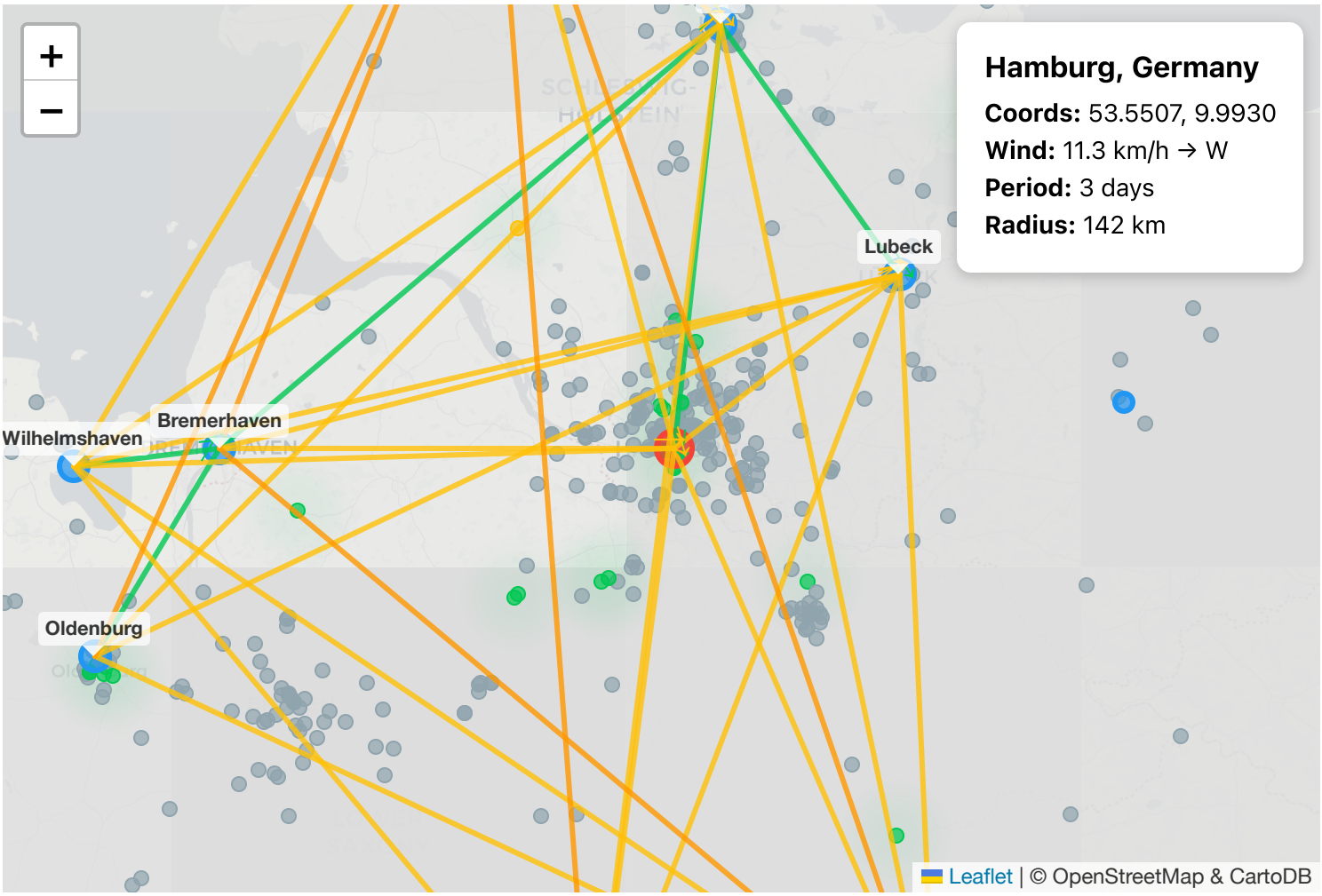

Air Quality Report — Hamburg, Germany

Key Insights

- **Highest PM2.5:** 15.2 µg/m³ (AQI 58, Moderate) at Kiel — 2026-03-15 06:00
- **Strongest front:** Kiel → Hamburg, Germany at 43 km/h S (correlation 89%, lag 2h)
- **Wind:** 11.3 km/h W — moderate — pollution disperses gradually
- **Assessment:** Moderate air quality. Generally safe, sensitive individuals may want to limit prolonged outdoor exertion.

Spikes

City	Time	PM2.5	Delta	Z-score
Hamburg, Germany	2026-03-14 06:00	14.1	+2.8	2.4
Lubeck	2026-03-15 02:00	10.2	+2.2	2.3
Lubeck	2026-03-15 03:00	13.4	+3.2	3.4

City	Time	PM2.5	Delta	Z-score
Kiel	2026-03-14 19:00	14.1	+2.2	2.2
Kiel	2026-03-15 06:00	15.2	+2.5	2.5
Bremerhaven	2026-03-15 02:00	10.1	+3.3	3.1
Bremerhaven	2026-03-15 05:00	14.1	+2.6	2.4
Bremerhaven	2026-03-15 22:00	8.9	+2.5	2.3
Bremerhaven	2026-03-15 23:00	11.4	+2.5	2.3
Wilhelmshaven	2026-03-15 05:00	13.2	+2.1	2.3
Wilhelmshaven	2026-03-15 23:00	9.4	+2.2	2.4
Oldenburg	2026-03-15 23:00	13.7	+4.5	4.0
Hannover	2026-03-14 19:00	11.7	+2.1	2.3
Hannover	2026-03-14 20:00	14.0	+2.3	2.6
Hannover	2026-03-15 20:00	11.2	+2.3	2.6
Wolfsburg	2026-03-14 23:00	10.7	+2.4	2.3
Wolfsburg	2026-03-15 18:00	8.8	+2.4	2.3
Flensburg	2026-03-15 07:00	14.9	+2.5	2.3

Pollution Fronts

From → To	Speed (km/h)	Direction	Lag (h)	Correlation
Kiel → Hamburg, Germany	43	S	2	89%
Wilhelmshaven → Bremerhaven	30	E	1	87%
Oldenburg → Bremerhaven	50	NE	1	86%
Bremerhaven → Kiel	17	NE	8	86%
Kiel → Lubeck	31	SE	2	85%
Wilhelmshaven → Hannover	8	SE	20	85%
Hannover → Wolfsburg	10	E	7	84%
Hamburg, Germany → Lubeck	19	NE	3	84%

From → To	Speed (km/h)	Direction	Lag (h)	Correlation
Wilhelmshaven → Kiel	18	NE	9	82%
Wilhelmshaven → Hamburg, Germany	12	E	10	82%
Hamburg, Germany → Hannover	44	S	3	81%
Wilhelmshaven → Wolfsburg	10	SE	22	80%
Hamburg, Germany → Wolfsburg	15	SE	9	80%
Kiel → Hannover	109	S	2	80%
Kiel → Wolfsburg	24	S	9	78%
Flensburg → Kiel	14	SE	5	76%
Lubeck → Wolfsburg	40	S	4	74%
Bremerhaven → Hamburg, Germany	12	E	8	72%
Flensburg → Hamburg, Germany	20	S	7	72%
Lubeck → Hannover	178	S	1	72%
Oldenburg → Wolfsburg	16	SE	12	72%
Wilhelmshaven → Lubeck	14	E	12	72%
Bremerhaven → Lubeck	18	E	8	71%
Oldenburg → Kiel	23	NE	8	71%
Oldenburg → Lubeck	23	NE	8	71%
Wilhelmshaven → Flensburg	41	NE	4	71%

Methodology

Data sources: Open-Meteo atmospheric model (~11km grid) + Sensor.Community citizen science network (517 sensors, 10 analysis nodes).

Spike detection: Z-score on hourly PM2.5 differences. A spike is flagged when the rate of change exceeds 2 standard deviations above the rolling mean — indicating a sudden pollution event rather than normal variation.

Front tracking: Cross-correlation analysis between city pairs with time lags from -24h to +24h. The lag at peak correlation reveals how many hours pollution takes to travel between two points. Combined with haversine distance, this gives front speed and direction.

Confidence: When both Open-Meteo (model) and Sensor.Community (ground sensors) agree on lag direction, confidence is boosted. Disagreement reduces confidence, highlighting uncertain fronts.

AQI Reference

AQI	Category	Health Guidance
0–50	Good	No restrictions
51–100	Moderate	Sensitive individuals: limit prolonged outdoor exertion
101–150	Unhealthy for Sensitive	Active children and adults: reduce prolonged outdoor exertion
151–200	Unhealthy	Everyone: reduce outdoor exertion
201–300	Very Unhealthy	Everyone: avoid outdoor activity
301–500	Hazardous	Stay indoors. Close windows.

Generated by airq — open source CLI air quality analyzer